

11 (a) period = 15 ms

C1

frequency ($= 1 / T$) = 67 Hz

A1 [2]

(b) zero

A1 [1]

(c) $I_{\text{r.m.s.}} = I_0 / \sqrt{2}$

C1

= 0.53 A

A1 [2]

(d) energy = $I_{\text{r.m.s.}}^2 \times R \times t$ or $\frac{1}{2} I_0^2 \times R \times t$
or

power = $I_{\text{r.m.s.}}^2 \times R$ and energy = power $\times t$

C1

energy = $0.53^2 \times 450 \times 30 \times 10^{-3}$

= 3.8 J

A1 [2]